ECON 1550 International Finance

A Tour of the World

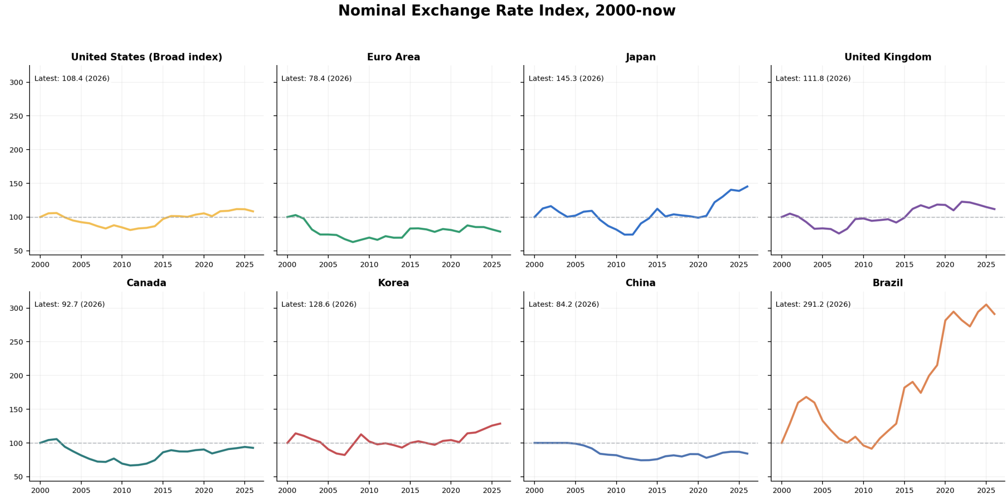
Agenda

- Long-run and recent exchange-rate behavior
- Aggregate stock-market performance and current yield curves
- Growth, inflation, and unemployment across major economies
- Where the post-2020 cycle looks global and where it does not

How to read the figures

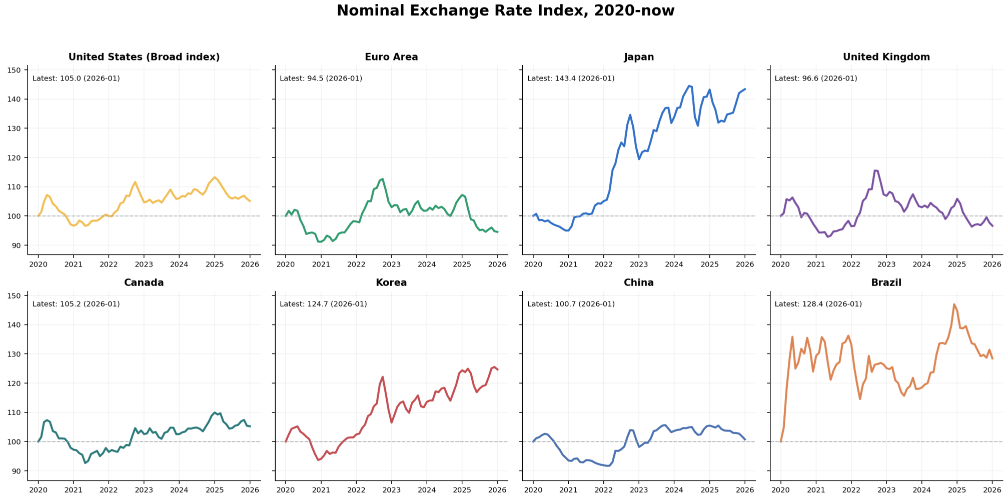
- Coverage: U.S., euro area, Japan, U.K., Canada, Korea, China, and Brazil.
- Exchange-rate and stock-market panels are indexed to 100 at the start of each window.
- Higher exchange-rate values always mean a stronger dollar; the U.S. panel uses a BIS broad dollar index.
- Recent macro slides combine IMF actual data and IMF outlook years.
- The yield-curve slide is a current snapshot; the euro-area panel uses Germany as a proxy.

Nominal exchange rates: 2000-now



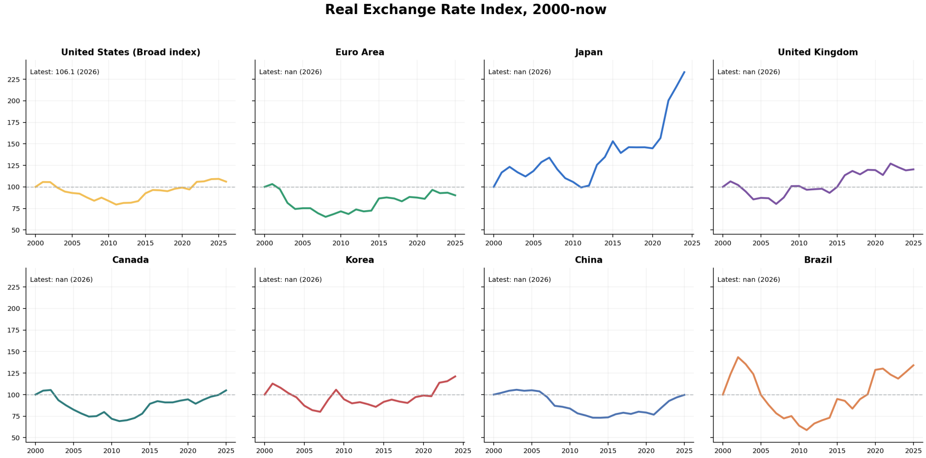
Source: FRED public monthly FX series. Non-U.S. panels plot local-currency units per dollar; the U.S. panel uses BIS broad dollar indexes.

Nominal exchange rates: 2020-now



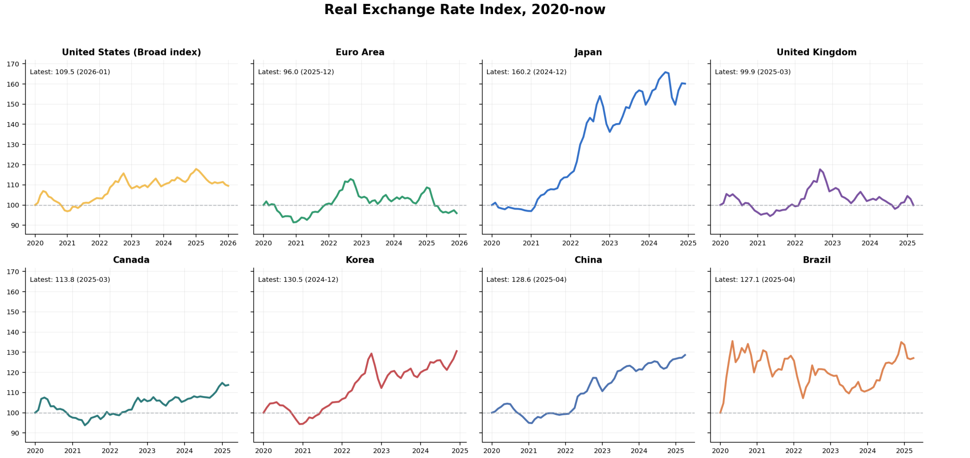
Source: FRED public monthly FX series. Non-U.S. panels plot local-currency units per dollar; the U.S. panel uses BIS broad dollar indexes.

Real exchange rates: 2000-now



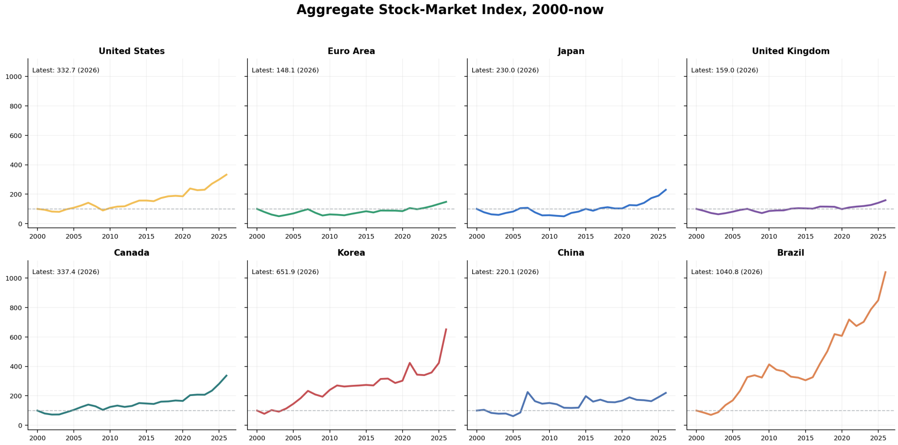
Source: FRED public monthly FX and CPI series. Japan and Korea use IMF-extended CPI tails through the latest actual inflation year because the FRED CPI endpoints stop early.

Real exchange rates: 2020-now



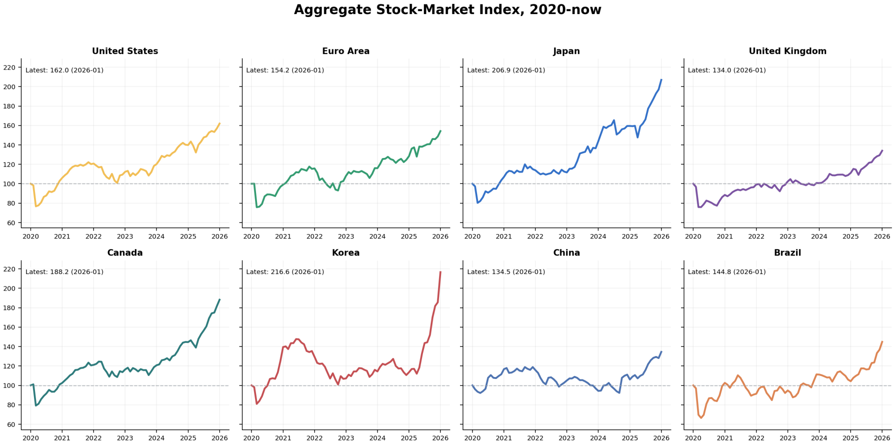
Source: FRED public monthly FX and CPI series. Japan and Korea use IMF-extended CPI tails through the latest actual inflation year because the FRED CPI endpoints stop early.

Aggregate stock-market performance: 2000-now



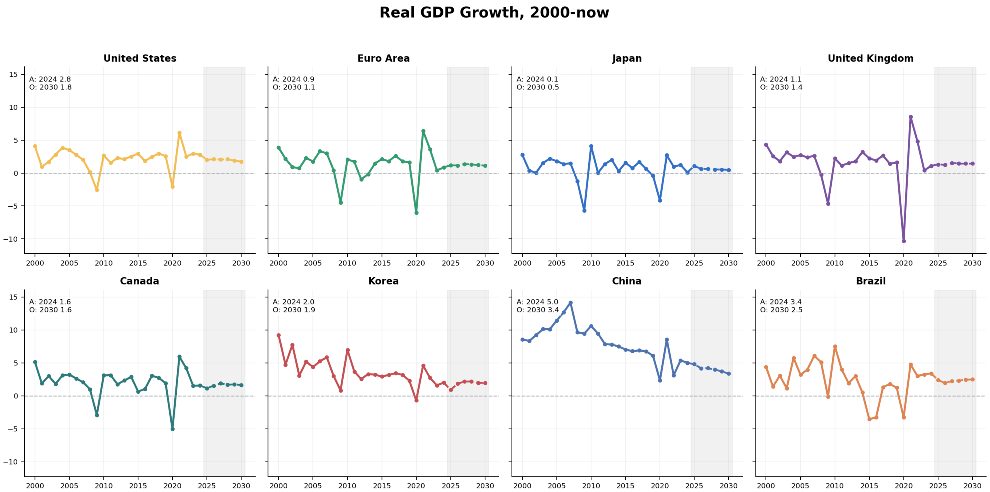
Source: FRED public OECD share-price indexes. Panels are indexed to 100 at the start of each window.

Aggregate stock-market performance: 2020-now



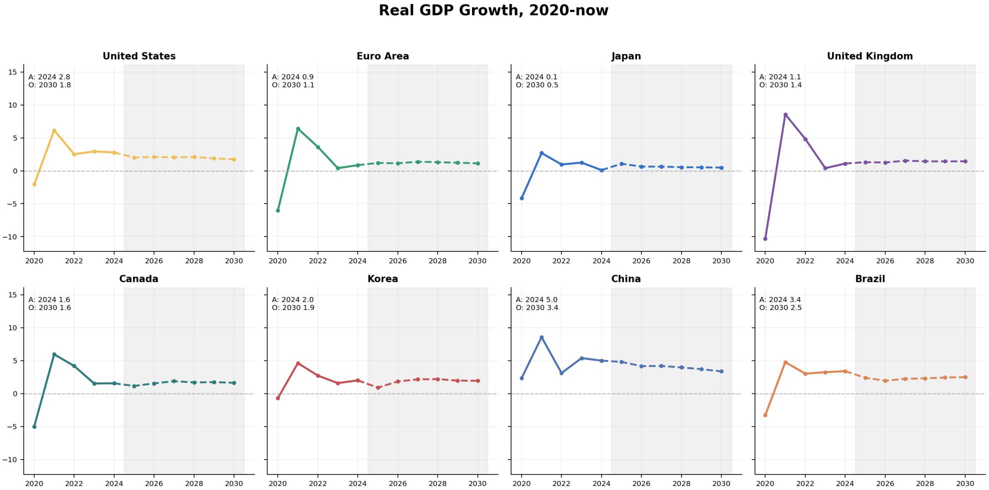
Source: FRED public OECD share-price indexes. Panels are indexed to 100 at the start of each window.

Real GDP growth: 2000-now



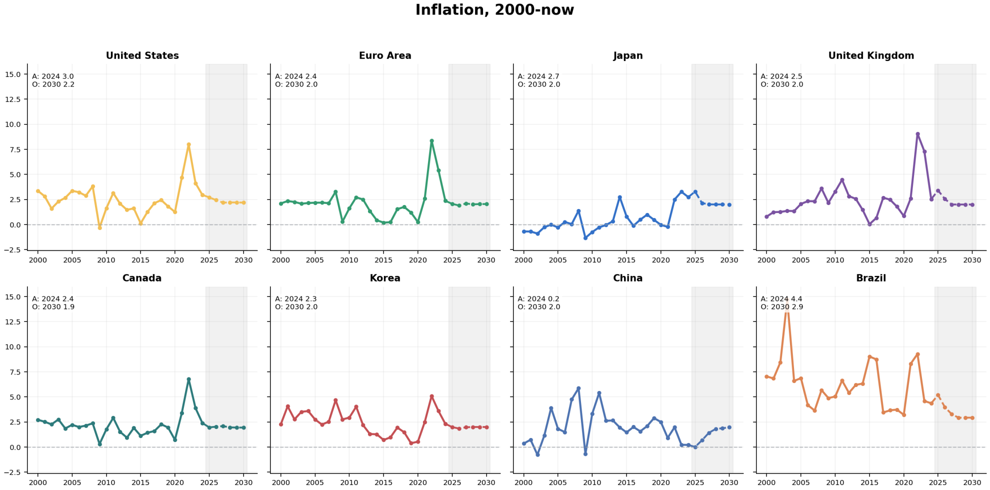
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Real GDP growth: 2020-now



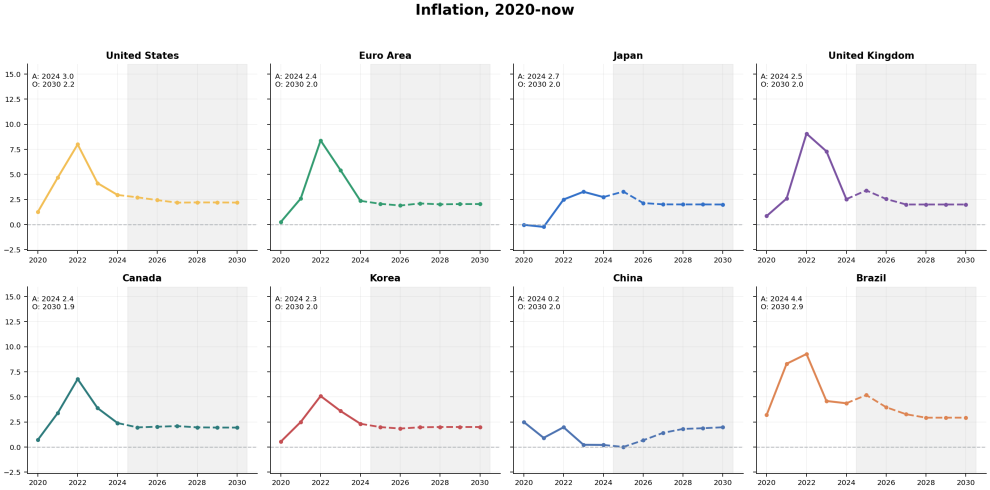
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Inflation: 2000-now



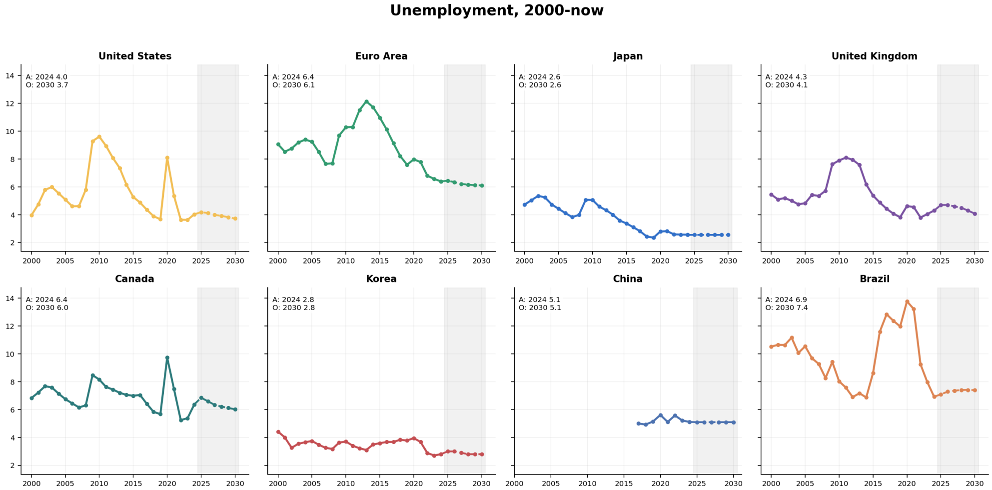
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Inflation: 2020-now



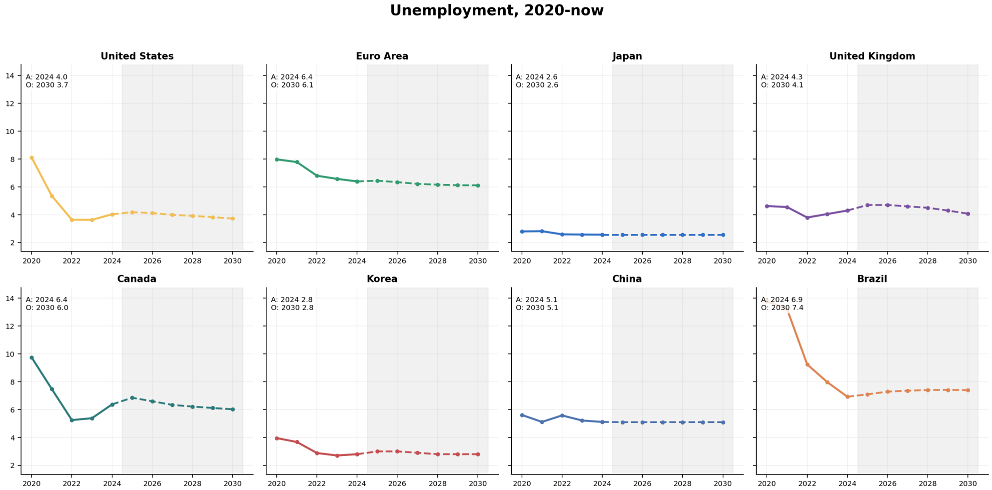
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Unemployment: 2000-now



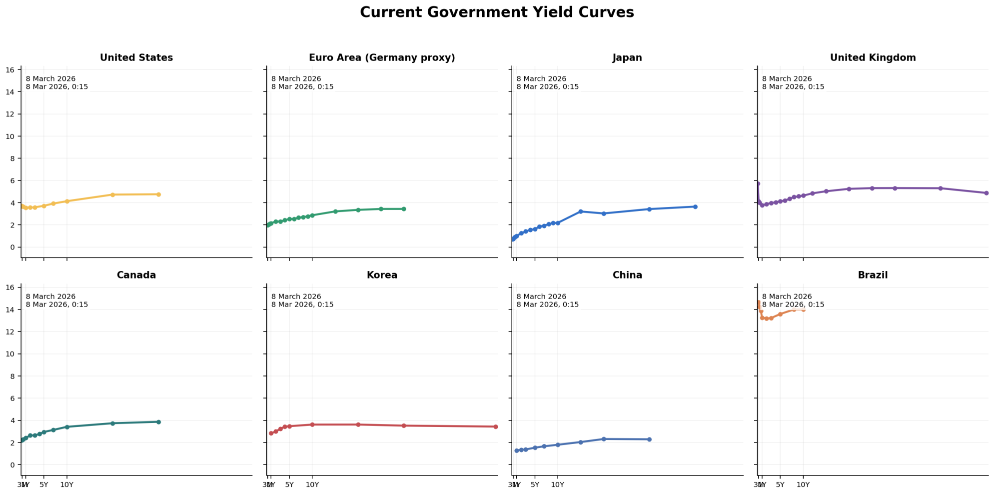
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Unemployment: 2020-now



Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Current yield curves



Source: WorldGovernmentBonds current country yield-curve pages. Snapshot date is shown in each panel; the euro-area panel uses Germany as a proxy.

Snapshot: 2020, latest actual, and latest IMF outlook

Real GDP Growth (%)

Unit	2020	A	O
US	-2.1	2.8	1.8
EA	-6.0	0.9	1.1
JP	-4.2	0.1	0.5
UK	-10.3	1.1	1.4
CA	-5.0	1.6	1.6
KR	-0.7	2.0	1.9
CN	2.3	5.0	3.4
BR	-3.3	3.4	2.5

A = latest actual, O = last WEO year

Inflation (%)

Unit	2020	A	O
US	1.3	3.0	2.2
EA	0.3	2.4	2.0
JP	-0.0	2.7	2.0
UK	0.9	2.5	2.0
CA	0.7	2.4	1.9
KR	0.5	2.3	2.0
CN	2.5	0.2	2.0
BR	3.2	4.4	2.9

A = latest actual, O = last WEO year

Unemployment (%)

Unit	2020	A	O
US	8.1	4.0	3.7
EA	8.0	6.4	6.1
JP	2.8	2.6	2.6
UK	4.6	4.3	4.1
CA	9.7	6.4	6.0
KR	4.0	2.8	2.8
CN	5.6	5.1	5.1
BR	13.8	6.9	7.4

A = latest actual, O = last WEO year

A = latest actual year in the October 2025 WEO release. O = the last year in that WEO horizon.

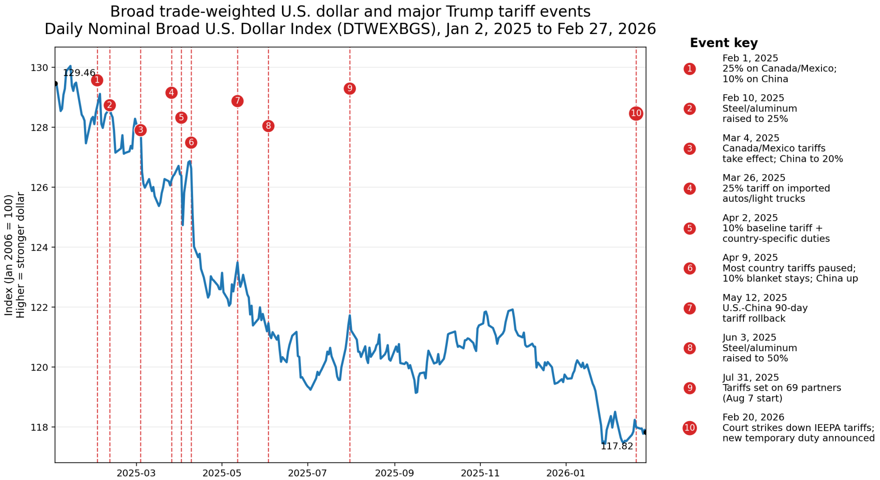
Source: IMF World Economic Outlook, October 2025. Gray shading marks estimate/projection years.

Bottom line to remember

The post-2020 cycle was global across the dollar, equity markets, inflation, labor markets, and the term structure, but the size and persistence of the response still varied a lot across countries.

Exchange rates and policy: current events

Trump tariffs and the broad dollar, 2025–2026



Timeline of major tariff events

	Date	Event
1	Feb 1, 2025	25% on Canada/Mexico; 10% on China
2	Feb 10, 2025	Steel/aluminum raised to 25%
3	Mar 4, 2025	Canada/Mexico tariffs take effect; China to 20%
4	Mar 26, 2025	25% tariff on imported autos/light trucks
5	Apr 2, 2025	10% baseline tariff + country-specific duties (“Liberation Day”)
6	Apr 9, 2025	Most country tariffs paused; 10% blanket stays; China up
7	May 12, 2025	U.S.–China 90-day tariff rollback
8	Jun 3, 2025	Steel/aluminum raised to 50%
9	Jul 31, 2025	Tariffs set on 69 partners (Aug 7 start)
10	Feb 20, 2026	Court strikes down IEEPA tariffs; new temporary duty announced

Sources: Reuters timeline of major tariff developments; White House fact sheet for the Feb. 20, 2026 temporary import duty.

The dollar fell despite rising tariffs

- Broad dollar index (DTWEXBGS): 129.46 on Jan 2, 2025 $\rightarrow$ 117.82 on Jan 27, 2026—roughly a 9% depreciation over 13 months.

- Standard textbook logic suggests tariffs should *strengthen* the dollar:
  - Tariffs reduce imports $\implies$ less demand for foreign currency.
  - Domestic production substitutes for imports $\implies$ higher demand for dollars.

- Yet the dollar weakened through most of the tariff escalation.

Why might tariffs weaken the dollar?

- Retaliation: trading partners imposed their own tariffs, reducing demand for U.S. exports and dollars.

- Uncertainty: frequent policy reversals (pause, rollback, escalation) may have reduced foreign appetite for U.S. assets.

- Growth expectations: tariff uncertainty can lower expected U.S. growth, making dollar assets less attractive.

- Institutional confidence: the IEEPA legal challenge (event 10) raised questions about the durability of the tariff regime.

- Bottom line: the exchange-rate effect of tariffs depends on the full general-equilibrium response, not just the direct trade-flow channel.

MFN averages vs. effective tariffs

- **MFN average tariff**: unweighted (or trade-weighted) mean across tariff lines that a country applies to all WTO members. Published annually by the WTO/ITC/UNCTAD.

- **Effective tariff**: what importers actually pay after special measures (Sections 232, 301), emergency surcharges (Section 122), tariff-rate quotas, and exclusions.

- The U.S. MFN average is low ($\approx$3.3%) and barely changed since 2016.

- But the *effective* tariff on many products is much higher in 2026, because the big changes came through **special measures layered on top of MFN duties**.

MFN tariff averages: 2016 (pre-Trump) vs. 2024

	2016 (pre-Trump)		2024 (latest WTO)	
Jurisdiction	Simple	Tr.-wtd	Simple	Tr.-wtd
United States	3.5	2.4	3.3	2.2
European Union	5.2	3.0	4.8	2.7
China	9.9	4.4	7.6	3.1
Canada	4.1	3.1	3.8	3.4
Mexico	7.0	4.5	7.4	5.4
Japan	4.0	2.1	3.7	1.9
United Kingdom	(in EU)		3.7	3.6
India	13.4	7.6	16.2	12.0
Brazil	13.5	10.4	12.0	8.0
South Korea	13.9	6.9	13.4	8.5
Vietnam	9.6	5.7	9.5	5.1

Source: WTO/ITC/UNCTAD tariff profiles. 2016 baseline from *World Tariff Profiles 2017*; 2024 from latest WTO tariff profile PDFs. Trade-weighted 2016 column uses 2015 imports. Simple = unweighted HS-6 average; Tr.-wtd = trade-weighted average. All figures are percentages.

U.S. effective tariff regime as of March 2026

- For many imports, the tariff bill stacks multiple layers:

$$\text{Effective duty} \approx \underbrace{\text{MFN duty}}_{\approx 3\%\text{ avg}} + \underbrace{10\%}_{\text{Sec. 122}} + \underbrace{\text{Sec. 301}}_{\text{China lists}} + \cdots$$

- **Section 122 surcharge** (Feb 24–Jul 24, 2026): 10% on all imports except certain critical minerals, energy products, and items already under Section 232.

- **Section 232**: steel/aluminum at 50%; autos/parts at 25%. These do *not* stack with the Section 122 surcharge.

Tariffs U.S. exports face in major markets

	Non-agricultural		Agricultural	
Partner	Simple	Tr.-wtd	Simple	Tr.-wtd
European Union	4.5	1.0	higher, varies	
Canada	2.5	1.9	higher, varies	
Mexico	5.6	3.5	higher, varies	
China	6.1	4.0	higher, varies	
United Kingdom	3.0	0.5	higher, varies	

- Agricultural tariffs are generally higher and more complex (TRQs, high peaks) across all partners.
- Retaliatory tariffs (targeted at U.S. products) are product-specific and time-varying—not captured by MFN averages.
- Partners with high MFN averages (India 16.2%, Korea 13.4%, Brazil 12.0%) impose high baseline barriers on U.S. exports even without retaliation.

Source: WTO tariff profile for the U.S. (2024), “Exports to major trading partners and duties faced” table; non-ag simple and trade-weighted averages

A 2026 yen episode that challenges simple UIP

- On January 23, 2026, the BOJ held its policy rate at 0.75% but sent a hawkish signal: upgraded growth forecasts, retained hawkish inflation outlook, said it would keep raising rates.

- Simple UIP with E_t in Krugman notation (USD per JPY):

$$E_t = E^e_{t+1} \, \frac{1 + R^Y_t}{1 + R_t}$$

- Holding E^e_{t+1} and R_t fixed, higher $R^Y_t \implies$ higher $E_t \implies$ yen appreciates.

What actually happened on January 23

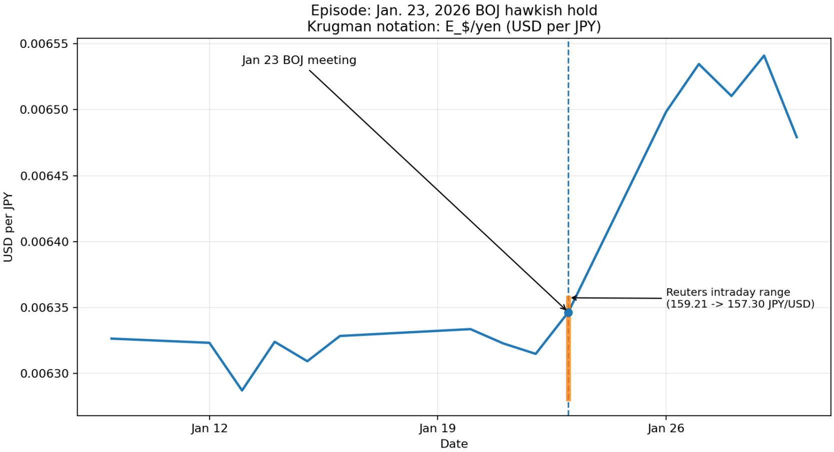
- During Gov. Ueda's press conference the yen *weakened*: the market quote rose to 159.21 JPY/USD.

- Only later did the yen reverse sharply, reaching about 157.30 JPY/USD.

- The initial move is the opposite of what simple UIP predicts.

- Daily data show net yen appreciation after the meeting, but miss the intraday sign reversal.

Daily exchange rate around the episode



Source: FRED DEXJPUS (daily JPY per USD), converted to USD per JPY. Orange segment: Reuters-reported intraday range (159.21 → 157.30 JPY/USD).

Why simple UIP gets the sign wrong

- The textbook comparative static in $E_t = E^e_{t+1}\,\frac{1+R^Y_t}{1+R_t}$ holds E^e_{t+1} and the risk premium fixed.
- In practice, all three moved at once on January 23:
  - Markets doubted how much tightening would actually follow.
  - The expected future exchange rate E^e_{t+1} shifted.
  - Intervention risk and fiscal-political constraints mattered.
- Once E^e_{t+1} and risk premia move, the spot rate can go the opposite way.

Bottom line to remember

Simple UIP captures the basic logic—more attractive foreign deposits should strengthen the foreign currency today—but the January 2026 BOJ episode shows that changing expectations, risk premia, and intervention risk can dominate, making the initial market reaction go the wrong way.

Appendix: Exact IDs (1/4)

United States: WEO code `USA`; nominal FX `NBUSBIS`; real FX `RBUSBIS`; CPI `CPIAUCNS`; stock `SPASTT01USM661N`; yield slug `united-states`.

Euro Area: WEO code `G163`; nominal FX `EXUSEU`, then invert; real FX uses formula on next slide with CPI `CP0000EZ19M086NEST`; stock `SPASTT01EZM661N`; yield slug `germany` (Germany proxy).

Japan: WEO code `JPN`; nominal FX `EXJPUS`; real FX uses formula on next slide with CPI `JPNCPIALLMINMEI`; stock `SPASTT01JPM661N`; yield slug `japan`.

United Kingdom: WEO code `GBR`; nominal FX `EXUSUK`, then invert; real FX uses formula on next slide with CPI `GBRCPIALLMINMEI`; stock `SPASTT01GBM661N`; yield slug `united-kingdom`.

Appendix: Exact IDs (2/4)

Canada: WEO code `CAN`; nominal FX `EXCAUS`; real FX uses formula on next slide with CPI `CANCPIALLMINMEI`; stock `SPASTT01CAM661N`; yield slug `canada`.

Korea: WEO code `KOR`; nominal FX `EXKOUS`; real FX uses formula on next slide with CPI `KORCPIALLMINMEI`; stock `SPASTT01KRM661N`; yield slug `south-korea`.

China: WEO code `CHN`; nominal FX `EXCHUS`; real FX uses formula on next slide with CPI `CHNCPIALLMINMEI`; stock `SPASTT01CNM661N`; yield slug `china`.

Brazil: WEO code `BRA`; nominal FX `EXBZUS`; real FX uses formula on next slide with CPI `BRACPIALLMINMEI`; stock `SPASTT01BRM661N`; yield slug `brazil`.

Appendix: Transformations (3/4)

1. FRED rule: download `https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES_ID` and use the monthly values exactly as returned.
2. Bilateral nominal FX is plotted as local-currency units per dollar: for Japan, Canada, Korea, China, and Brazil use the FRED value directly; for the euro area and U.K. set $E_t = 1/x_t$; for the U.S. use `NBUSBIS` directly.
3. Bilateral real FX for non-U.S. units is $q_t = E_t \times P_{US,t}/P_{i,t}$, with $P_{US,t} =$`CPIAUCNS` and $P_{i,t}$ from the IDs above; for the U.S. use `RBUSBIS` directly.
4. Long-run FX, real FX, and stock graphs: keep dates from 2000 onward; annualize by arithmetic mean of the 12 monthly observations in each year; plot $100 \times x_t/x_{2000}$.
5. Recent FX, real FX, and stock graphs: keep monthly observations with date $\geq$ 2020-01-01; plot $100 \times x_t/x_{base}$, where x_{base} is the first available observation in that window.
6. Macro graphs use raw annual percentages from WEO; no re-scaling, interpolation, seasonal adjustment changes, or other smoothing is applied.

Appendix: Exact Vintage And Adjustments (4/4)

1. IMF rule: use IMF WEO October 2025, exact raw file `overleaf/Data/imf/weo/weo_2025_10_v9_0_0.csv`. Indicator codes are `NGDP_RPCH` (real GDP growth), `PCPIPCH` (inflation), and `LUR` (unemployment). Exact WEO series code is `{country_code}.{indicator}.A`.
2. Projection rule for macro plots: `is_projection = year > LATEST_ACTUAL_ANNUAL_DATA`. Recent macro plots use a solid line through the latest actual year, a dashed line from that same point through the last WEO year, and gray shading from latest actual year $+ 0.5$ to last WEO year $+ 0.5$.
3. CPI extension rule if partner CPI ends before nominal FX ends: extend only through the latest actual inflation year and only with actual WEO inflation. If year y has annual inflation π_y, then monthly extension uses $P_{m+1} = P_m(1 + \pi_y/100)^{1/12}$. In this deck this affects Japan and Korea. No interpolation and no other extrapolation are used anywhere.
4. Yield-curve rule: open `https://www.worldgovernmentbonds.com/country/{slug}/`, extract `COUNTRY1`, POST it to `https://www.worldgovernmentbonds.com/wp-json/country/v1/main`, and plot the `Latest` series from `mainChart`. The exact cross section plotted in this PDF is `Resources/Data/yield_curve_current.csv`; all rows in this file have snapshot date 8 March 2026.
5. Exact transformed inputs used by the figures in this deck are `nominal_exchange_rate_monthly.csv`, `nominal_exchange_rate_annual.csv`, `real_exchange_rate_monthly.csv`, `real_exchange_rate_annual.csv`, `stock_market_monthly.csv`, `stock_market_annual.csv`, `macro_annual.csv`, and `yield_curve_current.csv` in `Resources/Data/`.